VACCINE DISTRIBUTION INVENTORY CONTROL SYSTEM AND COLD STORAGE THERMEL TRACKING HUB

PROJECT REPORT

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In partial fulfilment for the award of the degree of

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ABSTRACT

Vaccine distribution requires efficient inventory management, timely replenishment, batch and expiry tracking, and proper temperature monitoring. This project presents VaxTrack, a software-based vaccine distribution and inventory management system developed using Python, Flask, SQLite, and HTML/CSS. It provides separate dashboards for hospitals/pharmacies and manufacturers to manage vaccine stock, submit supply requests, and approve or reject orders.

The system includes CSV-based or simulated temperature-data analysis using Random Forest regression to forecast temperatures and identify potential cold-chain risks. It also provides rule-based stock indicators and redistribution suggestions. As an academic prototype, it uses synthetic data and is intended for workflow demonstration and software evaluation, not as a validated cold-chain safety system.

Keywords: Vaccine inventory, vaccine distribution, Flask, Random Forest regression, temperature forecasting, cold-chain monitoring.

TABLE OF CONTENTS

S.NO	TABLE OF CONTENTS	PAGE NO
	BONAFIDE CERTIFICATE	II
	DECLARATION	III
	ACKNOWLEDGEMENT	IV
	ABSTRACT	1
	LIST OF TABLES	7
	LIST OF FIGURES	8
	ABBREVIATIONS	9
1	**INTRODUCTION**	9
1.1	Objectives	9
1.2	Problem Identification	9
1.3	Outcome	10
1.4	Scope of the Project	11
1.5	Organisation of the Report	11
1.6	Significance of the Project	12
2	**LITERATURE SURVEY**	**13**
2.1	Review of Related Research Papers	13
2.2	Comparative Analysis of Existing Work	15
2.3	Research Gap and Positioning	17
2.4	Techniques Reviewed	17
3	**OBJECTIVES AND METHODOLOGY**	19
3.1	Introduction	19

3.2	Objectives of the Project	19
3.2.1	Vaccine Inventory Management	20
3.2.2	Batch and Expiry Tracking	20
3.2.3	Distribution Tracking	20
3.2.4	Temperature Data Analysis and Risk Identification	20
3.2.5	Temperature Prediction	21
3.2.6	Stock Status and Redistribution Support	21
3.2.7	Centralized Information Management	21
3.2.8	Simple and Accessible Interface	21
3.3	Scope of the Project	21
3.4	Proposed System	22
3.5	Methodology	22
3.5.1	Requirement Identification and System Design	23
3.5.2	Database Preparation and Module Development	23
3.5.3	Data Processing and Machine-Learning Integration	23
3.5.4	Result Display, Testing and Validation	23
3.6	System Architecture	24
3.6.1	Presentation Layer	24
3.6.2	Application Layer	24
3.6.3	Data Layer and Temperature-Analysis Component	24
3.7	Functional Requirements	25
3.8	Non-Functional Requirements	26
3.9	Database Design	26
3.9.1	Vaccine and Inventory Information	26
3.9.2	Batch Information	27
3.9.3	Distribution and Supply Request Information	27
3.9.4	Temperature Information and Data Relationships	27

3.10	Machine-Learning Methodology	27
3.10.1	Dataset Preparation and Feature Selection	27
3.10.2	Random Forest Regression	28
3.10.3	Model Training, Evaluation and Prediction	28
3.11	Temperature Limit Checking	29
3.12	Stock Status and Redistribution	30
3.13	System Workflow	30
3.14	Technologies Used	31
3.15	Hardware and Software Requirements	31
3.15.1	Hardware Requirements	31
3.15.2	Software Requirements	32
3.16	System Testing Methodology	32
3.16.1	Functional and Database Testing	32
3.16.2	Input Validation and Temperature Testing	32
3.16.3	Machine-Learning and User-Interface Testing	33
3.17	Expected Outcomes	33
3.18	Limitations and Assumptions	34
4	**PROPOSED WORK MODULES**	35
4.1	Introduction	35
4.2	Vaccine Inventory Management Module	35
4.3	Batch and Expiry Tracking Module	36
4.4	Vaccine Distribution Management Module	37
4.5	Temperature Data Analysis Module	38
4.6	Temperature Prediction Module	38
4.7	Stock Status and Redistribution Support Module	39
4.8	User Interface and Database Module	40
4.9	Module Integration	40

5	**RESULTS AND DISCUSSION**	42
5.1	Implemented Functional Scope	42
5.2	Vaccine Inventory and Batch Management	43
5.3	Temperature Forecasting and Cold-Chain Risk Analysis	43
5.4	Stock Status and Redistribution Suggestions	44
5.5	Supply Request and Manufacturer Workflow	44
5.6	Testing and Evaluation	45
5.7	Screenshot Evidence	46
5.8	Discussion	50
5.9	System Limitations	50
6	**CONCLUSIONS AND FUTURE WORK**	52
6.1	Conclusion	52
6.2	Future Enhancements	52
	REFERENCES	53
	APPENDICES	54

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LIST OF TABLES

Table No.	Name of the Table	Page No.
2.1	Comparative summary of existing work	15
3.1	Machine-Learning Process for Temperature Prediction	29
5.1	Inventory Module Evaluation	43
5.2	Temperature Model Evaluation	44
5.3	Supply Request Evaluation	45
5.4	Functional Test Cases	45
5.5	Cost Benefit Analysis	51

LIST OF FIGURES

Figure No.	Title of the Figure	Page No.
3.1	Overview of the Proposed VaxTrack System	25
4.1	System Architecture	37
4.2	VaxTrack Application Interface – Cold Storage and Thermal Tracking Hub	40
5.1	Dual-Portal Selection Interface	46
5.2	Hospital Dashboard	47
5.3	Cold Storage Thermal Tracking Hub	47
5.4	Shortage and Excess Stock Analysis	48
5.5	Vaccine Order Request Management	48
5.6	Manufacturer Operations Dashboard	49
5.7	Hospital Order Fulfillment and Distribution	49

ABBREVIATIONS

S. No	Abbreviation	Full Form
1	CSV	Comma-Separated Values
2	ML	Machine Learning
3	RF	Random Forest
4	MAE	Mean Absolute Error
5	MSE	Mean Squared Error
6	UI	User Interface
7	DB	Database
8	HTML	HyperText Markup Language
9	CSS	Cascading Style Sheets
10	SQL	Structured Query Language

CHAPTER – 1
INTRODUCTION

1.1. OBJECTIVES

The main objective of the Vaccine Distribution Inventory Control System and Cold Storage Thermal Tracking Hub (VaxTrack) is to develop a centralized web-based application for managing vaccine inventory, distribution requests, batch information, and temperature data.
The specific objectives of the project are:

- To develop a centralized system for maintaining vaccine inventory records.
- To manage vaccine batch details, available quantities, and expiry dates.
- To provide a structured workflow for submitting and processing vaccine supply requests.
- To analyse uploaded or simulated temperature datasets.
- To apply Random Forest regression for temperature forecasting.
- To identify potentially concerning temperature values by comparing readings and forecasts with configured limits.
- To display stock-status indicators and provide rule-based redistribution suggestions.
- To improve the accessibility and organization of vaccine distribution information through a role-based web application.

1.2. PROBLEM IDENTIFICATION

Vaccine distribution requires proper inventory management, accurate batch records, timely supply requests, and careful review of storage-temperature information. When these activities are maintained using separate spreadsheets or manual registers, it may become difficult to identify stock shortages, monitor expiry dates, and coordinate supply requirements.

Limited visibility of inventory information can make it harder for users to determine which vaccines are available and which batches require attention. Similarly, manually reviewing temperature records may make it difficult to identify trends or estimate possible future temperature changes.

The project addresses these information-management challenges by developing VaxTrack, a software-based system that combines inventory management, batch tracking, supply request processing, and temperature-data analysis within a common interface.

The temperature-analysis component uses uploaded CSV files or clearly labelled simulated readings. It applies data preprocessing and Random Forest regression to demonstrate temperature forecasting. The system does not use physical temperature sensors or provide live IoT monitoring. Therefore, it is intended for academic demonstration and software evaluation rather than validated vaccine-safety monitoring.

1.3. OUTCOME

The expected outcome of the project is a web-based software prototype that demonstrates the organized management of vaccine inventory, distribution requests, and temperature information.

The system is designed to provide the following capabilities, depending on the features implemented and tested:

- Maintenance and display of vaccine inventory records.
- Organization of vaccine batch details and expiry information.
- Submission and review of supply requests by hospital/pharmacy and manufacturer users.
- Storage and analysis of uploaded or simulated temperature records.
- Temperature forecasting using Random Forest regression when the model is successfully trained and integrated.
- Display of temperature risk indicators based on configured limits.
- Stock-status information and illustrative redistribution suggestions based on predefined rules.
- Role-based access to relevant application functions.

The project demonstrates how web application development, database management, and machine-learning techniques can be combined in a single workflow. Its actual performance must be reported using completed functional tests and measured model-evaluation results. Simulated temperature data cannot establish the safety of real vaccine storage conditions.

1.4. SCOPE OF THE PROJECT

The scope of VaxTrack covers the development of a web-based application for vaccine inventory management, batch and expiry tracking, supply request processing, and temperature-data analysis.

The application uses Python and Flask for backend development, SQLite for data storage, and HTML and CSS for the user interface. Pandas and NumPy support data processing, while scikit-learn provides the Random Forest regression functionality for temperature forecasting.
The project includes the following areas:

- Maintaining vaccine inventory and batch information.
- Recording and reviewing expiry dates.
- Submitting and processing supply requests through role-based workflows.
- Uploading CSV temperature datasets or using clearly labelled simulated readings.
- Preprocessing temperature data and generating forecasts.
- Comparing temperature readings or forecasts with configured limits.
- Displaying stock indicators and rule-based redistribution suggestions.

The project does not include physical temperature sensors, real-time IoT monitoring, GPS tracking, RFID integration, or automatic control of cold-storage equipment. Demand indicators and redistribution suggestions are based on predefined rules rather than trained demand-prediction or logistics-optimization models.

The system is an academic software prototype. Further development, security testing, and validation using suitable real-world data would be required before considering operational healthcare deployment.

1.5. ORGANISATION OF THE REPORT

This report is organised into six chapters to explain the problem, proposed solution, methodology, implementation, evaluation, and future development of VaxTrack.

- **Chapter 1 – Introduction:** Presents the objectives, problem identification, outcome, scope, organisation of the report, and significance of the project.
- **Chapter 2 – Literature Survey:** Reviews related research on vaccine inventory management, vaccine supply chains, cold-chain temperature monitoring, and temperature forecasting. It also identifies the research gap and discusses the techniques reviewed.

- **Chapter 3 – Methodology:** Explains the proposed methodology, system architecture, algorithms, requirements, database design, dataset preparation, temperature forecasting approach, and evaluation metrics.
- **Chapter 4 – Result and Discussion / Proposed Work Modules:** Describes the system modules, workflows, interfaces, implementation approach, and the outputs produced by the developed application.
- **Chapter 5 – Conclusion and Future Enhancements:** Summarises the project outcomes and discusses possible improvements to inventory management, forecasting, security, and future integration.
- **Chapter 6 – References:** Lists the research papers, technical documentation, and other sources referred to in the report.

1.6. SIGNIFICANCE OF THE PROJECT

VaxTrack demonstrates the potential of a centralized software application to organize vaccine inventory and distribution information. Maintaining inventory, batch, and expiry details in a structured database can make relevant records easier to access and review.

The supply request workflow provides a common process for hospital/pharmacy users to request supplies and for manufacturer users to review and process those requests. Stock indicators and rule-based redistribution suggestions demonstrate how existing inventory information can support preliminary operational decisions.

The temperature-analysis module provides an academic example of processing time-series data and applying Random Forest regression to estimate future temperature values. Comparing readings or forecasts with configured limits can help demonstrate how potentially concerning values may be highlighted for further review.

The project is also significant from an educational perspective because it combines web development, database operations, data preprocessing, and machine learning in one application. However, its outputs depend on the quality of the input data and the correctness of the implemented rules and model. The prototype does not replace professional cold-chain procedures and does not establish that vaccines have been stored safely.

CHAPTER – 2
LITERATURE SURVEY

Vaccine distribution and inventory management involve maintaining stock availability, tracking batch and expiry information, coordinating supply requests, and reviewing storage-temperature records. Previous studies have examined vaccine supply-chain management, cold-chain monitoring, inventory planning, and machine-learning techniques. These studies provide a foundation for developing the Vaccine Distribution Inventory Control System and Cold Storage Thermal Tracking Hub (VaxTrack).

Dixit et al. [1] investigated capacity management under uncertain demand and different distribution policies in the Indian vaccine supply chain. Their study used modelling approaches to examine the relationship between capacity decisions, distribution policies, costs, and service levels. The research highlights the importance of coordinating demand and supply. In contrast, VaxTrack currently provides stock-based indicators rather than a trained demand-forecasting model or a capacity-optimization system.

Dey et al. [2] reviewed optimization models for pandemic vaccine supply-chain management. Their study considered topics such as inventory management, allocation, distribution logistics, and transportation planning. It demonstrates the complexity of developing optimized vaccine-distribution strategies. VaxTrack uses simpler rule-based stock indicators and redistribution suggestions; it does not implement the mathematical optimization methods discussed in the review.

Jiang, Jia, and Guo [3] proposed an Internet of Things (IoT)-enabled framework for sustainable vaccine cold-chain management. Their work describes hardware and software components for temperature monitoring and the management of vaccine-related information. This provides a useful comparison for cold-chain system design. VaxTrack differs because it is a software-only prototype and does not use physical sensors, IoT devices, or direct refrigerator connectivity.

Sripada et al. [4] developed a decision-support framework for vaccine distribution across a multi-tier cold-chain network. Their approach uses mathematical optimization models to consider inventory, ordering, transportation, personnel, and shortage costs. This work demonstrates the value of structured decision support in vaccine distribution. VaxTrack

provides a more limited academic implementation that organizes inventory and supply requests without claiming globally optimal distribution.

Fahrni et al. [5] reviewed the management of COVID-19 vaccine cold-chain logistics. Their study discussed the challenges involved in maintaining suitable storage and transportation conditions for vaccines. It demonstrates the importance of organized cold-chain procedures and temperature information. VaxTrack addresses the software-analysis aspect by allowing temperature records to be uploaded and reviewed, without claiming to provide physical or live temperature monitoring.

Kartoglu and Ames [6] examined the role of temperature monitoring in maintaining vaccine quality throughout the cold chain. Their review emphasizes the importance of monitoring temperature conditions during storage and transportation. This supports the inclusion of temperature-data analysis in VaxTrack. However, the project's software-generated indicators are intended for demonstration and do not replace approved temperature-monitoring equipment or professional assessment.

Das et al. [7] studied temperature integrity in vaccine storage devices at different levels in three Indian states. Their research used electronic temperature data loggers to examine temperature exposure in healthcare facilities. The findings illustrate why accurate temperature records are important in vaccine storage management. Unlike that field study, VaxTrack uses uploaded CSV files or simulated readings and does not collect measurements from physical equipment.

Duijzer, van Jaarsveld, and Dekker [8] presented a literature review of vaccine supply-chain management. Their study examined logistical challenges and identified research opportunities in vaccine distribution. It highlights the importance of considering inventory, supply, and distribution as connected activities. VaxTrack follows this general principle by bringing inventory records and supply request workflows together in one application.

Hanson et al. [9] reviewed evidence concerning vaccine exposure to freezing temperatures in the cold chain. Their study identified continued concerns about temperatures falling below recommended ranges. This research supports the need to review temperature records and identify possible deviations. In VaxTrack, such deviations are highlighted by comparing available readings or forecasts against configured temperature limits.

Pedregosa et al. [10] introduced scikit-learn, a Python machine-learning library that supports supervised and unsupervised learning algorithms. The library provides tools for implementing and evaluating machine-learning models. It supports the technical implementation of Random Forest regression in VaxTrack, where the model is used to forecast temperature values from available historical data. The reliability of the resulting forecasts must be assessed using suitable evaluation data.

2.1 COMPARATIVE SUMMARY OF EXISTING WORK

The following table compares ten research papers and technical publications relevant to vaccine supply-chain management, cold-chain monitoring, inventory planning, and machine learning. The comparison identifies the approaches used in existing work and explains their relevance to the proposed VaxTrack system.

2.1: Comparative Summary of Existing Work

Ref.	Author(s) and Year	Paper Title	Approach / Methodology	Remarks
[1]	Dixit et al. (2025)	Capacity Management Under Uncertain Demand and Distribution Policies: The Case of Indian Vaccine Supply Chain	Demand modelling and capacity-management analysis	Highlights supply and capacity planning challenges; VaxTrack does not implement this optimization approach.
[2]	Dey et al. (2024)	Optimization Modeling for Pandemic Vaccine Supply Chain Management: A Review and Future Research Opportunities	Review of vaccine inventory, allocation, and distribution optimization models	Provides background on advanced optimization; VaxTrack uses rule-based suggestions instead.
[3]	Jiang et al. (2024)	Internet of Things (IoT)-Enabled Framework for a Sustainable Vaccine Cold Chain Management System	IoT-based cold-chain monitoring framework	Uses hardware and connectivity that are not part of the current software-only project.

[4]	Sripada et al. (2023)	A Decision Support Framework for Optimal Vaccine Distribution Across a Multi-Tier Cold Chain Network	Mathematical optimization for vaccine distribution	Demonstrates advanced distribution planning beyond the scope of the current prototype.
[5]	Fahrni et al. (2022)	Management of COVID-19 Vaccines Cold Chain Logistics: A Scoping Review	Review of cold-chain logistics and management practices	Supports the importance of cold-chain procedures and organized monitoring.
[6]	Kartoglu and Ames (2022)	Ensuring Quality and Integrity of Vaccines Throughout the Cold Chain: The Role of Temperature Monitoring	Review of vaccine temperature-monitoring practices	Establishes the importance of temperature records; VaxTrack provides software-based analysis only.
[7]	Das et al. (2020)	Temperature Integrity and Exposure of Vaccines to Suboptimal Temperatures in Cold Chain Devices at Different Levels in Three States of India	Field study using electronic temperature data loggers	Demonstrates temperature deviations in real storage environments; VaxTrack does not use physical sensors.
[8]	Duijzer et al. (2018)	Literature Review: The Vaccine Supply Chain	Review of vaccine logistics and supply-chain research	Highlights the need to consider connected vaccine supply-chain activities.
[9]	Hanson et al. (2017)	Is Freezing in the Vaccine Cold Chain an Ongoing Issue? A Literature Review	Review of studies on vaccine exposure to freezing temperatures	Highlights the importance of identifying potentially unsuitable temperature conditions.
[10]	Pedregosa et al.	Scikit-learn: Machine Learning in Python	Machine-learning library and	Provides the software foundation for

	(2011)		algorithm implementation	implementing and evaluating Random Forest regression.

2.2 RESEARCH GAP AND POSITIONING

Existing studies focus on vaccine inventory management, cold-chain monitoring, demand forecasting, and supply-chain optimization. However, VaxTrack integrates inventory management, batch and expiry tracking, supply requests, temperature analysis, and redistribution suggestions in a single web application.

The system uses uploaded CSV files or simulated temperature readings and Random Forest regression for temperature forecasting. Stock indicators and redistribution suggestions are based on predefined rules. Since the project does not use physical sensors, RFID, GPS, or IoT devices, it is intended for academic demonstration rather than real-world vaccine safety validation.

2.3 TECHNIQUES REVIEWED

Inventory Management and Batch Tracking

Maintains vaccine quantities, batch details, and expiry dates in a database to support stock monitoring.

Supply Request Management

Allows hospitals and pharmacies to submit vaccine requests and manufacturers to review and process them.

Rule-Based Stock and Redistribution Analysis

Uses predefined rules to identify potential stock shortages or excess stock and suggest redistribution.

Temperature Data Preprocessing

Prepares uploaded or simulated temperature records for analysis by checking and organizing the available data.

Random Forest Regression

Uses an ensemble of decision trees to forecast temperature values from historical records.

Model Evaluation Using MAE and RMSE

Uses MAE and RMSE to measure prediction errors when suitable test data is available.

Temperature-Limit Checking

Compares recorded or forecast temperatures with configured limits and highlights potential deviations.

Role-Based Web Application

Uses Flask and a database to provide separate functions for hospital/pharmacy and manufacturer users.

CHAPTER – 3
OBJECTIVES AND METHODOLOGY

3.1 INTRODUCTION

The Vaccine Distribution Inventory Control System and Cold Storage Thermal Tracking Hub, known as VaxTrack, is a software-based system designed to support vaccine inventory management, distribution tracking, batch monitoring, and temperature data analysis. Proper management of vaccines is important because vaccines must be stored and distributed carefully to maintain their quality.

Manual inventory management can make it difficult to track stock quantities, batch details, expiry dates, and distribution records. Similarly, reviewing temperature records manually can make it difficult to identify unusual readings and possible temperature-related risks. These challenges highlight the need for a centralized system that organizes vaccine-related information and presents it in an understandable format.

VaxTrack provides a web-based interface through which users can manage inventory details, monitor batch information, record distribution activities, and analyse temperature data. The system also incorporates a machine-learning approach using the Random Forest algorithm for temperature prediction, where suitable historical or prepared data is available.

The proposed system uses Python and Flask for backend development, HTML and CSS for the user interface, and SQLite for data storage. Pandas and NumPy support data processing, while scikit-learn is used for machine-learning operations.

This chapter describes the objectives, scope, proposed system, architecture, methodology, requirements, database design, temperature analysis, inventory management, and testing approach of VaxTrack.

3.2 OBJECTIVES OF THE PROJECT

The main objective of VaxTrack is to develop a centralized application for managing vaccine inventory, supply requests, batch details, and temperature-related information. The system aims to improve the organization and accessibility of vaccine distribution records.

The specific objectives are to:

- Maintain vaccine inventory details and available stock quantities.
- Track vaccine batches and their expiry dates.
- Support distribution records and supply request processing.
- Provide a workflow for manufacturers to review and process requests.
- Analyse uploaded or simulated temperature records.
- Demonstrate temperature forecasting using a machine-learning approach, where implemented and evaluated.
- Compare recorded or forecast temperature values with configured limits.
- Identify possible stock shortages and excess stock using predefined rules.
- Present inventory and temperature information through a simple web interface.
- Support better review of vaccine-related information through centralized records.

These objectives define the intended functionality of the system. Actual implementation and performance claims must be based on the features and tests completed in the project

3.2.1 VACCINE INVENTORY MANAGEMENT

The system aims to maintain organized records of vaccine names, available quantities, batch numbers, and other relevant inventory details. This helps users view stock information without depending entirely on separate manual records.

3.2.2 BATCH AND EXPIRY TRACKING

The system supports the recording and viewing of batch details and expiry dates. This information helps users identify batches that are approaching their expiry dates and prioritize appropriate stock review.

3.2.3 DISTRIBUTION TRACKING

The system organizes distribution-related records and supply requests. These records help users review the movement of vaccine stock between the participating roles represented in the application.

3.2.4 TEMPERATURE DATA ANALYSIS

The system processes temperature readings provided through uploaded files or simulated datasets. The readings can be examined to identify values that fall outside the temperature limits configured for the application.

3.2.5 TEMPERATURE PREDICTION

A Random Forest regression model is included as a machine-learning approach for estimating temperature values from suitable input features. Its usefulness depends on the availability, quality, and preparation of the training data.

3.2.6 STOCK STATUS AND REDISTRIBUTION SUPPORT

Predefined rules are used to identify low-stock or excess-stock conditions and provide illustrative redistribution suggestions. These suggestions support manual review; they are not generated by a trained demand-prediction model or a logistics-optimization algorithm.

3.2.7 CENTRALIZED INFORMATION MANAGEMENT

The application brings inventory, batch, request, and temperature-related information into one system. This reduces the need to refer to multiple separate records when reviewing the information maintained by the prototype.

3.2.8 SIMPLE AND ACCESSIBLE INTERFACE

The web interface is designed to present information through forms, tables, dashboards, and status indicators. The aim is to make the main functions understandable and convenient for users.

3.3 SCOPE OF THE PROJECT

The scope of VaxTrack covers software-based vaccine inventory management, distribution record maintenance, and temperature data analysis.

The inventory module covers the recording and viewing of vaccine stock details, batch information, expiry dates, and stock status. The distribution module supports the maintenance of distribution-related records. These features help organize information and make it easier to review vaccine movement.

The temperature analysis module accepts data from an uploaded CSV file or a simulated dataset. The application processes the readings and compares them with configured temperature limits. A Random Forest regression model can be used to estimate temperature values when the required data and trained model are available.

The system is developed as an academic software prototype. It does not include physical temperature sensors, direct IoT connectivity, or automatic real-time temperature collection. Its temperature analysis depends on the data supplied to the application.

The project focuses on organizing information and providing decision-support features. It does not replace professional cold-chain procedures, official vaccine storage guidelines, or the judgement of authorized healthcare personnel.

3.4 PROPOSED SYSTEM

The proposed VaxTrack system combines inventory management, distribution tracking, and temperature data analysis within a single web-based application.
Users interact with the system through a web interface. Requests are handled by the Flask backend, which processes the submitted information and communicates with the database. Inventory and distribution records are maintained for later retrieval and review.

For temperature analysis, the system accepts uploaded or simulated readings. The data is checked and processed before being used for temperature-limit comparison or prediction. The results are displayed to help users identify readings that require attention.

The proposed system uses two types of decision-support logic. The first is **rule-based checking**, which compares stock quantities or temperature readings against configured conditions. The second is **machine-learning-based prediction**, which estimates temperature values using a Random Forest regression model when suitable trained data is available.

The combination of these features provides a structured way to review vaccine-related information. However, the output is intended for decision support and must not be treated as a substitute for verified storage conditions or official safety procedures.

3.5 METHODOLOGY

The methodology describes the steps followed to design and develop the VaxTrack application. The work is divided into requirement identification, system design, database preparation, module development, data processing, machine-learning integration, and testing.

3.5.1 REQUIREMENT IDENTIFICATION AND SYSTEM DESIGN

The initial stage involves identifying the main requirements of vaccine inventory management, batch tracking, supply request processing, and temperature analysis. The system is designed around two principal user groups: hospital or pharmacy users and manufacturer users.

The application is divided into modules so that each function can be developed and tested separately. The web interface handles user input and displays information, the Flask application processes requests and applies business rules, and the database stores the required records.

3.5.2 DATABASE PREPARATION AND MODULE DEVELOPMENT

The database is designed to store inventory details, batch information, supply requests, and temperature records as required by the implemented features. Related records are organized to support retrieval and updates.

The main modules are developed using Python and Flask. HTML and CSS are used to create the web pages, while Flask routes connect user actions to the corresponding application functions. Database operations are used to create, retrieve, update, and process stored information.

3.5.3 DATA PROCESSING AND MACHINE-LEARNING INTEGRATION

Temperature data supplied through CSV files is checked and prepared before analysis. Processing may include verifying required columns, converting values into appropriate data types, handling missing entries, and organizing records in time order.

Where the Random Forest regression model is implemented, prepared historical data is used to train the model and generate forecasts. The model's performance should be measured using a separate test dataset where appropriate. If the model or evaluation has not been completed, it should be reported as work in progress rather than as a verified feature.

3.5.4 RESULT DISPLAY, TESTING AND VALIDATION

The results are presented through the application's interface using tables, status indicators, and relevant summaries. Stock conditions are determined using predefined rules, while temperature indicators are based on comparisons with configured limits.

Testing checks whether the implemented modules accept valid inputs, handle invalid inputs appropriately, store and retrieve information correctly, and display results as expected. The system's actual performance is documented using the results obtained during testing.

3.6 SYSTEM ARCHITECTURE

The system architecture explains how the different components of VaxTrack communicate with one another. The application follows a layered structure consisting of the presentation layer, application layer, data layer, and analysis component.

3.6.1 PRESENTATION LAYER

The presentation layer provides the user interface. HTML and CSS are used to organize the pages and present the available functions. Users interact with the application through forms, tables, and result displays, depending on the implemented features.

3.6.2 APPLICATION LAYER

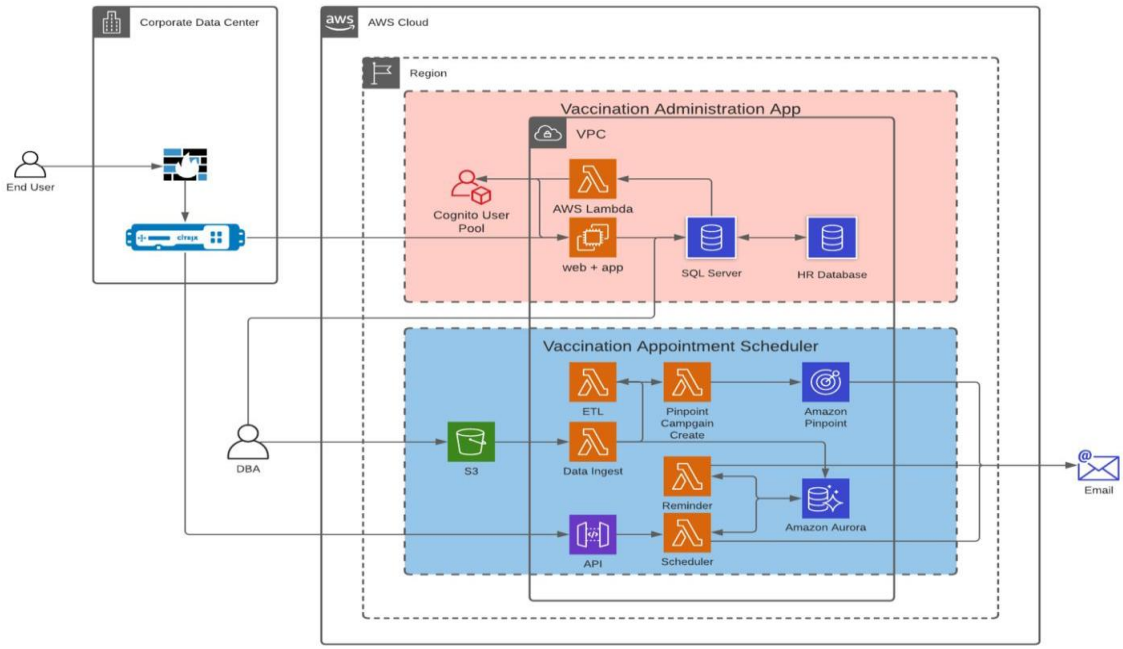
The application layer is implemented using Python and Flask. It receives user requests, validates input, executes the required operations, and coordinates communication between the interface and database.

This layer also connects the relevant functions for inventory management, distribution records, and temperature analysis.

3.6.3 DATA LAYER AND TEMPERATURE-ANALYSIS COMPONENT

The data layer uses SQLite to store the records required by the application. Flask-SQLAlchemy can be used to manage database operations through application models.

The temperature-analysis component processes uploaded or simulated readings and generates the results supported by the implemented analysis method. If a trained model is used, its output is displayed alongside the configured temperature limits. The component does not receive live readings from physical sensors.



3.1: Overview of the Proposed VaxTrack System

3.7 FUNCTIONAL REQUIREMENTS

Functional requirements describe the operations that the application is expected to perform.

The inventory management function maintains vaccine stock details and allows users to review available quantities. Batch management records relevant batch information, including expiry dates, so that users can identify stock requiring attention. Distribution-related functions organize supply records and provide a basis for reviewing vaccine movement.

The supply request function allows hospital or pharmacy users to submit requests. Manufacturer users can review submitted requests and approve or reject them according to the implemented workflow.

The temperature-data function accepts supported input data, such as CSV files or simulated readings. The application processes the available records and presents the analysis results. Temperature-limit checking compares recorded or forecast values with configured limits and identifies potential deviations.

The stock-status function applies predefined rules to identify possible low-stock and excess-stock conditions. The result-display function presents the information through tables,

dashboards, and status indicators. Only functions implemented and tested in the final application should be described as completed.

3.8 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the expected qualities of the application.

Usability: The interface should be straightforward to navigate, with clear labels and understandable result displays.

Reliability: The application should handle valid records consistently and provide appropriate feedback when an operation fails.

Data integrity: Records should be validated before storage to reduce incomplete or incorrect information.

Maintainability: The code should be organized into understandable modules so that individual functions can be updated or corrected.

Performance: Common operations, such as retrieving inventory records and processing small datasets, should complete within a reasonable time in the intended development environment.

Security: Input validation and appropriate access restrictions should be considered when handling application data. A prototype should not be assumed to provide production-level security without suitable implementation and testing.

Scalability: The modular design can support future enhancements, although larger deployments may require changes to the database, infrastructure, and application architecture.

3.9 DATABASE DESIGN

The database is responsible for maintaining structured information used by the application. SQLite is selected because it provides a lightweight database suitable for an academic software prototype.

The database structure should reflect the information actually stored by the application. The following logical entities describe the main information categories.

3.9.1 VACCINE AND INVENTORY INFORMATION

This information contains details such as vaccine name, available quantity, and other inventory attributes supported by the implementation. It allows users to retrieve stock information and review availability.

3.9.2 BATCH INFORMATION

Batch records may contain a batch identifier, vaccine reference, batch number, expiry date, and quantity associated with the batch. These fields help distinguish one batch from another.

3.9.3 DISTRIBUTION AND SUPPLY REQUEST INFORMATION

Supply request records contain the information needed to identify a request, the requesting organization, the requested vaccine or quantity, and its processing status. These records support the workflow between hospital or pharmacy users and manufacturer users.

Distribution records, where implemented, organize the relevant details of vaccine supply and movement.

3.9.4 TEMPERATURE INFORMATION AND DATA RELATIONSHIPS

Temperature records contain the available time and temperature information required for analysis. Uploaded data may be processed for display or forecasting without representing live measurements.

The database structure should maintain appropriate relationships between the records used by the application. The exact table names, columns, and relationships should match the database models in the final source code.

3.10 MACHINE-LEARNING METHODOLOGY

Machine learning is included in VaxTrack to support temperature prediction from suitable historical or prepared data. The selected approach is Random Forest regression, which combines the predictions of multiple decision trees to produce a numerical estimate.

3.10.1 DATASET PREPARATION AND FEATURE SELECTION

The temperature dataset is obtained from a CSV upload or a clearly identified simulated dataset. The input data is inspected to identify required columns, missing values, invalid entries, and inconsistencies. Temperature records are organized appropriately before they are used for analysis.

The features used by the model should be selected from the fields actually available in the dataset. The target variable is the temperature value to be predicted. Any time-based features or previous temperature values must be derived from available records using a consistent procedure.

Simulated readings are useful for demonstrating the workflow but do not represent measurements from a real vaccine refrigerator.

3.10.2 RANDOM FOREST REGRESSION

Random Forest regression is an ensemble learning method that combines predictions from multiple decision trees. Each tree is trained using a sample of the training data and a subset of candidate features. For regression, the predictions from the trees are combined to produce the final estimated value.

The method can model relationships between input features and temperature values. However, prediction quality depends on the amount and quality of training data, the suitability of the features, and the similarity between the training data and the data to which the model is applied. The model should be described as implemented only if the project's source code actually trains and uses it.

3.10.3 MODEL TRAINING, EVALUATION AND PREDICTION

Where the model is implemented, the prepared dataset is divided into training and testing portions using a suitable procedure. The training data is used to fit the model, while the test data is used to evaluate its predictions. For time-series data, the split should preserve chronological order where appropriate to reduce information leakage from future records.

Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) may be used to measure prediction error. MAE measures the average absolute difference between actual and predicted values. RMSE gives greater weight to larger errors.

The measured values of MAE and RMSE, dataset size, and train/test sizes should be included only after actual evaluation. The model's predictions can then be presented through the web interface for review. If training and evaluation remain incomplete, these procedures should be described as intended work rather than completed results.

3.1 - Machine-Learning Process for Temperature Prediction

S. No.	PROCESS	DESCRIPTION
1	Data Collection	Obtain temperature data from uploaded or simulated datasets.
2	Data Preprocessing	Check missing values and invalid records.
3	Feature Selection	Select suitable input variables for prediction.
4	Model Training	Train the Random Forest regression model using available training data.
5	Model Evaluation	Compare predicted values with actual values using suitable metrics.
6	Temperature Prediction	Estimate temperature values for suitable new inputs.
7	Result Display	Display the prediction results in the application.

3.11 TEMPERATURE LIMIT CHECKING

Temperature-limit checking is used to identify recorded or forecast values that fall outside the configured range. The application compares each available temperature value with the minimum and maximum limits configured for the relevant analysis.

If a value is outside the configured range, the application can display a warning or risk indicator. Values within the configured range can be displayed as being within the selected limits. These labels are based on the rules applied by the software.

The analysis depends on the correctness of the input data and the configured limits. A warning does not independently establish that a vaccine has become unsafe, and a normal reading does not prove that the vaccine has remained within acceptable conditions throughout storage and transport.

Because VaxTrack uses uploaded or simulated readings, the module demonstrates software-based analysis rather than real-time cold-chain monitoring.

3.12 STOCK STATUS AND REDISTRIBUTION

The stock-status module helps users review inventory quantities and identify potential shortages or excess stock. It uses predefined rules to classify stock conditions according to the thresholds or conditions configured in the application.

For example, a vaccine quantity below a specified threshold may be identified as low stock, while a quantity above a selected threshold may be identified as excess stock. The actual conditions must match the rules implemented in the source code.

The redistribution function can provide illustrative suggestions based on the available stock information. These suggestions may help users identify locations or records that require further review. They do not automatically calculate an optimal redistribution plan and should not be described as the output of a trained demand-forecasting model unless such a model is actually implemented.

The module is intended to support inventory review and decision-making rather than replace the judgment of authorized personnel.

3.13 SYSTEM WORKFLOW

The VaxTrack workflow begins when a user opens the web application and accesses the relevant workflow. Hospital or pharmacy users review inventory and batch details, maintain available records, and submit supply requests when additional stock is needed. Manufacturer users review incoming requests and process them through the available approval or rejection functions.

For temperature analysis, the user provides a supported CSV dataset or uses clearly labelled simulated data. The application checks and prepares the available records before applying the implemented analysis method. Where forecasting is available, the output is compared with configured temperature limits and displayed for review.

The stock-status module applies predefined conditions to identify possible shortages and excess stock. Redistribution suggestions are presented as rule-based guidance. Users can review the displayed information to understand the current records and determine whether further action is needed.

The overall workflow connects inventory management, request processing, and temperature-data analysis within one application. Its outputs depend on the information entered and the functions implemented.

3.14 SOFTWARE AND TECHNOLOGIES USED

The project uses software tools and libraries that support web development, data storage, data processing, and machine learning.

Python: Python is used for backend logic, data processing, and machine-learning operations.

Flask: Flask provides the web application framework for handling requests and connecting application functions.

HTML: HTML defines the structure of the web pages and forms.

CSS: CSS controls the presentation and layout of the user interface.

SQLite: SQLite stores structured application records in a lightweight database.

Pandas: Pandas supports reading, organizing, and processing tabular data, including CSV files.

NumPy: NumPy supports numerical operations used in data processing.

Scikit-learn: Scikit-learn provides machine-learning functionality, including Random Forest regression and relevant model-evaluation tools.

These technologies support the development of a modular prototype. Only libraries actually used in the final implementation should be included in the final report.

3.15 HARDWARE AND SOFTWARE REQUIREMENTS

3.15.1 HARDWARE REQUIREMENTS

The prototype can be developed and executed on a standard computer capable of running Python and a web browser. A dedicated sensor device is not required for the current software-only implementation.

The basic hardware requirements include:

- A desktop or laptop computer.
- Sufficient memory to run the development environment and application.
- Adequate storage for source code, database files, and datasets.
- A keyboard, mouse, and display for development and testing.

The required resources may vary depending on dataset size and the development environment.

3.15.2 SOFTWARE REQUIREMENTS

The software requirements include:

- An operating system that supports Python and the selected development tools.
- Python for application development.
- Flask for web application functionality.
- SQLite for database storage.
- A supported web browser for accessing the application.
- Pandas and NumPy for data processing, where used.
- Scikit-learn for Random Forest regression, where implemented.
- A code editor or integrated development environment.

The required Python packages should be installed in the project environment before execution. The application should be tested using the versions specified in the project setup.

3.16 SYSTEM TESTING METHODOLOGY

Testing is performed to verify that the application behaves as expected under normal and invalid input conditions. It also helps identify issues in data processing, database operations, and result presentation.

3.16.1 FUNCTIONAL AND DATABASE TESTING

Functional testing checks inventory operations, batch and expiry record handling, supply request submission, and manufacturer-side request processing. Each function is tested using appropriate inputs to verify that the expected action takes place.

Database testing checks whether records are stored, retrieved, and updated correctly. It also checks whether the displayed information corresponds to the saved records and whether invalid data is handled appropriately.

3.16.2 INPUT VALIDATION AND TEMPERATURE TESTING

Input validation testing checks how the application responds to missing fields, invalid quantities, incorrect data formats, and unsupported file uploads. The application should display an appropriate response rather than process invalid input as valid information.

Temperature testing checks whether supported datasets can be processed and whether the configured limits are applied correctly. Test cases should include values within the range and values below or above it. The observed output should be compared with the expected output.

3.16.3 MACHINE-LEARNING AND USER-INTERFACE TESTING

Where a forecasting model is implemented, testing checks whether it can be trained using the prepared dataset and produce predictions for supported inputs. MAE and RMSE should be calculated from actual test predictions if model evaluation is completed. No performance values should be assumed or inserted without measurement.

User-interface testing checks navigation, form submission, tables, dashboard displays, and the presentation of status indicators. The application should be reviewed in the supported browser to identify layout or usability problems.

The test report should record the test case, input, expected result, observed result, and pass/fail status. Only tests actually performed should be marked as passed.

3.17 EXPECTED OUTCOMES

The expected outcome of VaxTrack is a working software prototype that organizes vaccine inventory and distribution information and supports temperature data analysis.
The system is intended to provide the following benefits:

- Centralized access to available vaccine inventory records.
- Organized storage of batch and expiry information.
- Easier review of distribution-related records.
- Identification of temperature readings outside configured limits.
- Temperature estimates through the Random Forest model when the model is successfully trained and integrated.
- A structured view of stock status and possible redistribution needs.
- Reduced dependence on scattered manual records for the information handled by the application.
-

The actual outcomes must be described according to the features implemented and tested. The system should not be presented as a validated cold-chain monitoring solution or as proof that vaccine quality has been maintained.

3.18 LIMITATIONS AND ASSUMPTIONS

The current prototype has several limitations that must be considered when interpreting its outputs.

First, temperature information is obtained from uploaded or simulated datasets. The system does not collect readings directly from physical temperature sensors and does not provide live IoT monitoring.

Second, the usefulness of temperature prediction depends on the quality and quantity of the available data. A model trained on limited or simulated data may not perform reliably in real-world conditions.

Third, temperature-limit checking depends on the limits configured for the application. Those limits must be selected using appropriate guidance rather than assumed to apply to every vaccine.

Fourth, the inventory and distribution results depend on the accuracy of the information entered or imported by users. Incomplete records may produce misleading stock summaries.

Fifth, redistribution support is based on available information and simple rules. It does not necessarily account for every operational factor involved in real vaccine distribution.

Finally, the project is an academic prototype. Further development, security testing, integration with reliable monitoring equipment, and evaluation using real operational data would be required before considering deployment in an actual vaccine supply chain.

CHAPTER – 4
PROPOSED WORK MODULES

4.1 INTRODUCTION

The Vaccine Distribution Inventory Control System and Cold Storage Thermal Tracking Hub (VaxTrack) is designed to support the management of vaccine inventory, batch information, distribution records, and temperature data. The system combines these functions in a web-based application to provide organized access to vaccine-related information.

Vaccine distribution requires proper stock management, accurate batch records, and careful review of temperature conditions. When these activities are managed using separate records, it may become difficult to retrieve information, identify stock shortages, track expiry dates, and review temperature variations. VaxTrack aims to address these information-management challenges through a centralized software system.

The proposed application is developed using Python and Flask for backend operations, HTML and CSS for the user interface, and SQLite for database storage. Pandas and NumPy support data processing, while scikit-learn provides the Random Forest regression functionality for temperature prediction when a suitable trained model is available.

The system is divided into individual modules. Each module handles a specific activity, and the Flask backend connects these functions. The following sections explain the purpose, operation, and importance of each module.

4.2 VACCINE INVENTORY MANAGEMENT MODULE

The vaccine inventory management module maintains information about the vaccines available in the system. Inventory records help users understand the quantity of stock available and review vaccine details whenever required.

The module supports the organization of inventory information, such as vaccine names and quantities. Additional details, including batch references and expiry information, may be associated with inventory records according to the database design.

When inventory information is entered or updated, the backend processes the request and stores the supported information in the SQLite database. Users can retrieve the stored records through

the web interface. This arrangement reduces dependence on scattered records and makes inventory information easier to review.

The module also supports stock-status monitoring. When stock thresholds are configured, the available quantity can be compared against the threshold to identify possible low-stock conditions. Such information can help users plan stock reviews and determine whether further action may be required.

Accurate inventory management depends on the information entered into the system. Therefore, quantities and other fields should be validated before records are saved. The module provides a structured approach to maintaining vaccine stock information but does not automatically guarantee that physical stock matches the database.

4.3 BATCH AND EXPIRY TRACKING MODULE

The batch and expiry tracking module supports the recording and review of batch-level vaccine information. A batch number helps distinguish one group of vaccine stock from another and provides a reference when reviewing related records.

Expiry dates are important for stock management because vaccines must be handled according to their applicable storage and expiry requirements. By maintaining expiry information, the system can help users identify batches that are approaching their recorded expiry dates.

The module organizes batch information so that users can retrieve and review the relevant details. Where expiry-status checking is implemented, the application can compare the recorded expiry date with the current date and identify records that require attention.

Batch information can also support distribution record management. When a distribution transaction is associated with a batch, users can review which batch was involved in that transaction, provided the relevant relationship is maintained in the database.

This module improves the organization of batch-related information and supports appropriate stock review. However, the accuracy of the results depends on the correctness of the batch numbers, dates, and quantities entered into the application.

4.4 VACCINE DISTRIBUTION MANAGEMENT MODULE

The vaccine distribution management module maintains information about vaccine stock movement. Distribution records help users review how much stock has been distributed and examine the available details of each transaction.

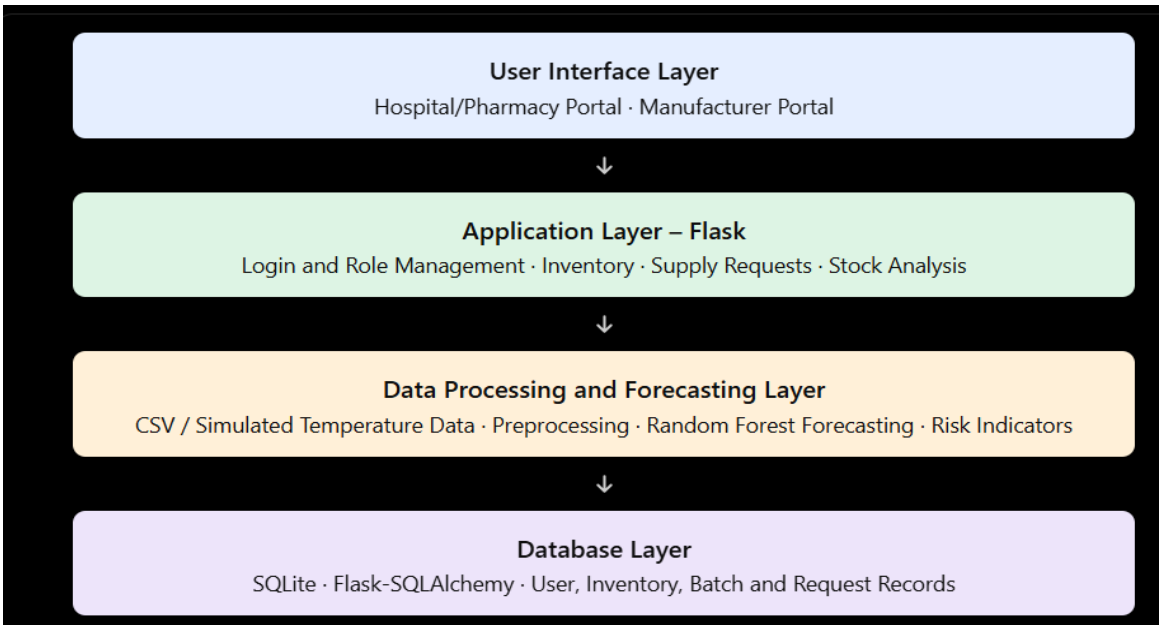
The module can store relevant information supported by the application, such as vaccine details, quantities, destination information, and distribution dates. These records provide a structured history of distribution activities.

The backend receives distribution-related requests and processes them according to the implemented application logic. The resulting information is stored in or retrieved from the database and presented through the user interface.

Distribution records can be considered alongside inventory details when reviewing stock availability. For example, recorded distribution quantities may help explain changes in available stock when the inventory and transaction logic are correctly connected.

Maintaining distribution information can improve visibility of stock movement and support record review. However, the application depends on users entering complete and accurate transaction details. The module should therefore be considered an information-management tool rather than an automatic logistics or delivery-tracking system.

4.1: SYSTEM ARCHITECTURE



4.5 TEMPERATURE DATA ANALYSIS MODULE

The temperature data analysis module processes temperature information provided through uploaded CSV files or simulated datasets. It is intended to help users review recorded temperature values and identify readings that fall outside the configured limits.

The first stage involves accepting the temperature dataset and checking whether the required data is available. The input may be examined for missing values, invalid numerical entries, and inconsistent formats. Pandas can be used to read and organize tabular data, while NumPy can support numerical operations where required.

After preprocessing, the temperature readings can be compared with the limits configured for the application. Readings below the lower limit or above the upper limit can be flagged for review. Readings within the configured range can be identified as falling within those specified limits.

The results can be presented in a table or chart to make temperature patterns easier to understand. A timestamp, when available in the dataset, can help users relate readings to their recorded time. Such visual representations can help identify variations in the supplied data.

The module uses uploaded or simulated temperature information. It does not directly collect live readings from physical sensors in the current prototype. Furthermore, an out-of-range flag indicates a reading that requires attention; it does not independently establish whether a vaccine has been damaged or is safe for use.

4.6 TEMPERATURE PREDICTION MODULE

The temperature prediction module incorporates a Random Forest regression approach to estimate temperature values from suitable input features. This module demonstrates the application of machine learning to temperature data analysis.

Random Forest regression works by building multiple decision trees and combining their predictions to produce a numerical estimate. This approach can represent relationships between input features and a target value, including certain nonlinear relationships.

The process begins with preparing a suitable dataset. The required input features and temperature target must be identified, and unsuitable records must be handled before training. When sufficient data is available, the dataset can be divided into training and testing portions.

The training dataset is used to fit the model. The testing dataset can then be used to evaluate predictions against actual temperature values. Metrics such as Mean Absolute Error and Root Mean Squared Error may be used to measure prediction error, provided the model has actually been trained and evaluated.

After training, the model can estimate temperature values for suitable new inputs. These predictions can be displayed through the application to support the review of temperature data. The reliability of this module depends on the quantity, quality, and relevance of the available data. Predictions based on limited or simulated data may not represent real-world storage conditions. Therefore, model outputs should be treated as estimates and should not replace direct temperature measurements or established vaccine-handling procedures.

4.7 STOCK STATUS AND REDISTRIBUTION SUPPORT MODULE

The stock-status and redistribution support module helps users review available vaccine quantities and identify potential stock-management needs. It uses the inventory information maintained by the application.

Where stock thresholds are configured, the system can compare recorded quantities with those thresholds and flag possible low-stock conditions. This information helps users recognize when stock may require further review.

Batch and expiry details can also contribute to stock-management decisions. Users can examine available quantities alongside expiry dates to identify records that need attention. Such information can support appropriate stock rotation and review procedures.

Where the application maintains location-level inventory information, users may compare the stock available at different locations to identify possible redistribution opportunities. These comparisons can provide useful information for planning, but redistribution decisions require additional consideration of demand, transport arrangements, storage conditions, expiry dates, and applicable guidelines.

The module provides basic decision support based on the information available in the system. It does not guarantee optimized distribution or automatically account for all real-world operational constraints

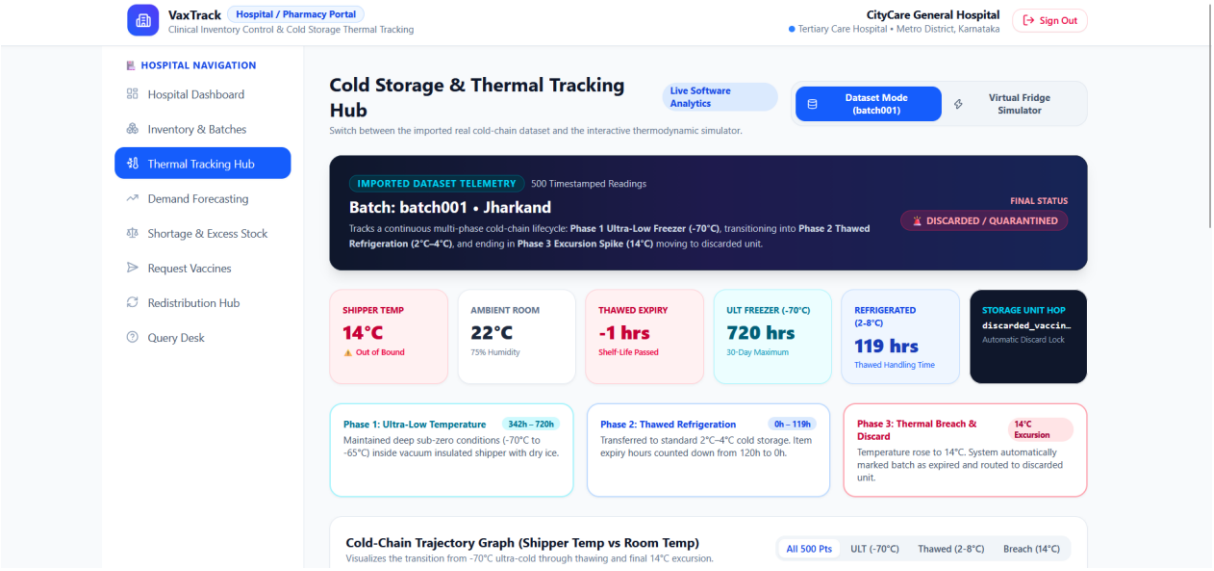
4.8 USER INTERFACE AND DATABASE MODULE

The user interface and database module connect the user-facing pages with the information stored and processed by the application. The interface is developed using HTML and CSS, while Flask manages the backend requests and application operations.

The interface presents the functions available to users through pages, forms, tables, and result displays. Users can interact with the supported inventory, batch, distribution, and temperature-analysis features through these pages.

SQLite is used to maintain structured application records. When a user submits information, the backend validates the request and performs the appropriate database operation. When information is requested, the backend retrieves the relevant records and returns them for display.

Separating the interface, backend processing, and database operations makes the application easier to organize and maintain. It also supports future improvements, such as adding additional reports or extending the existing inventory functions.



4.2: VaxTrack Application Interface – Cold Storage and Thermal Tracking Hub

4.9 MODULE INTEGRATION

The modules work together through the Flask backend to provide a unified application. The inventory and batch modules maintain vaccine stock information, while the distribution module

records vaccine movement. These records can be retrieved from the database when users need to review the available information.

The temperature analysis module processes uploaded or simulated readings and compares them with configured limits. When a trained Random Forest model is available, the prediction module can generate estimates from suitable input features.

The backend coordinates these operations and returns the results to the user interface. This structure allows users to access the functions through a common application rather than managing each activity in a completely separate system.

The quality of the integrated results depends on accurate inventory records, appropriate temperature data, and correct implementation of the application logic. Each module must therefore be tested individually and as part of the complete application.

CHAPTER – 5
RESULTS AND DISCUSSION

This chapter presents the functional scope, testing approach and evaluation of the AI-Based Vaccine Distribution and Inventory Management System (VaxTrack). The system integrates vaccine inventory management, batch and expiry tracking, supply request processing, temperature-data analysis and rule-based stock indicators through a web application.

The application is developed using Python, Flask, Flask-SQLAlchemy, SQLite, HTML, CSS, pandas, NumPy and scikit-learn. Its temperature-analysis module processes uploaded CSV data or clearly labelled simulated readings and uses Random Forest regression to forecast temperature values. The forecasts are compared with configured temperature limits to display an indicator for further review.

VaxTrack is a software-only academic prototype. It does not use physical temperature sensors, ESP32, RFID, GPS or live IoT monitoring. Therefore, its results demonstrate software functionality and model behaviour on the evaluated dataset, not real-world vaccine safety.

5.1 IMPLEMENTED FUNCTIONAL SCOPE

The project brings several vaccine-management activities together in one application. The main functional areas are:

- **Role-based access:** Provides separate application workflows for hospital/pharmacy users and manufacturer users.
- **Inventory and batch management:** Stores vaccine stock details, batch numbers, quantities and expiry dates.
- **Supply request management:** Allows hospital/pharmacy users to submit requests and manufacturer users to review and respond to them.
- **Temperature analysis:** Accepts uploaded CSV data or simulated readings for preprocessing and temperature forecasting.
- **Stock status and redistribution:** Uses predefined rules to identify stock conditions and present illustrative redistribution suggestions.
- **Query handling:** Provides a facility for users to submit queries through the application.

The final report should include only features verified in the running application. Any incomplete feature should be identified as work in progress rather than a completed result.

5.2 VACCINE INVENTORY AND BATCH MANAGEMENT

The inventory module is designed to maintain vaccine stock information in a structured database. Batch details and expiry dates help users review available stock and identify batches approaching expiry. Stock-status indicators provide a quick overview based on the rules implemented in the application.

The reliability of these indicators depends on the accuracy and completeness of the entered records. They support information review but do not replace physical stock verification or approved vaccine-handling procedures.

5.1: Inventory Module Evaluation

Function	Expected behaviour	Actual result
Add vaccine stock	Valid stock details are saved	To be recorded after testing
Maintain batch details	Batch information is stored and displayed	To be recorded after testing
Display expiry information	Stored expiry dates are visible	To be recorded after testing
Display stock status	Implemented stock rules produce the relevant indicator	To be recorded after testing

5.3 TEMPERATURE FORECASTING AND COLD-CHAIN RISK ANALYSIS

The temperature-analysis module demonstrates how historical temperature data can be processed and used for short-term forecasting. The input is provided through a CSV file or generated simulated data; it is not collected directly from a refrigerator.

The processing workflow includes validating the required timestamp and temperature fields, preparing time-series features, training a Random Forest regression model and generating temperature forecasts. The predicted values are compared with configured limits to produce an indicator for further review.

Model performance should be evaluated using suitable holdout data. Mean Absolute Error (MAE) measures the average absolute difference between predicted and observed temperatures. Root Mean Squared Error (RMSE), if calculated, gives greater weight to prediction errors.

5.2: Temperature Model Evaluation

Evaluation item	Result to record
Dataset source	Uploaded dataset name or simulated dataset
Number of input rows	Enter actual count
Number of valid rows	Enter actual count
Training and holdout sizes	Enter actual counts
MAE (°C)	Enter measured value
RMSE (°C), if calculated	Enter measured value
Forecast generation	Record the actual test outcome
Configured-limit indicator	Record the actual test outcome

5.4 STOCK STATUS AND REDISTRIBUTION SUGGESTIONS

VaxTrack provides stock-status information and illustrative redistribution suggestions using predefined rules. These features are intended to help users review potential stock shortages or excess stock when the required inventory information is available.

The current functionality should not be described as a trained demand-forecasting model or an optimized logistics solution. The usefulness of the suggestions depends on the stock records and the rules configured in the application.

The final evaluation should verify that the stock overview displays the expected indicators and that any redistribution suggestions are consistent with the available records.

5.5 MANUFACTURER-SIDE SUPPLY COORDINATION

The manufacturer workflow is intended to provide a structured method for reviewing supply requests submitted by hospital/pharmacy users. Depending on the implemented functions, a manufacturer user can view pending requests and approve or reject them.

5.3: Supply Request Evaluation

Test scenario	Expected behaviour	Actual result
Submit a valid supply request	The request is recorded	To be recorded after testing
View submitted requests	The relevant request is displayed	To be recorded after testing
Approve a request	The status changes as implemented	To be recorded after testing
Reject a request	The status changes as implemented	To be recorded after testing

5.6 SYSTEM TESTING AND PERFORMANCE EVALUATION

Functional testing should be conducted to verify the major application workflows. The following test plan provides a structure for recording the observed results.

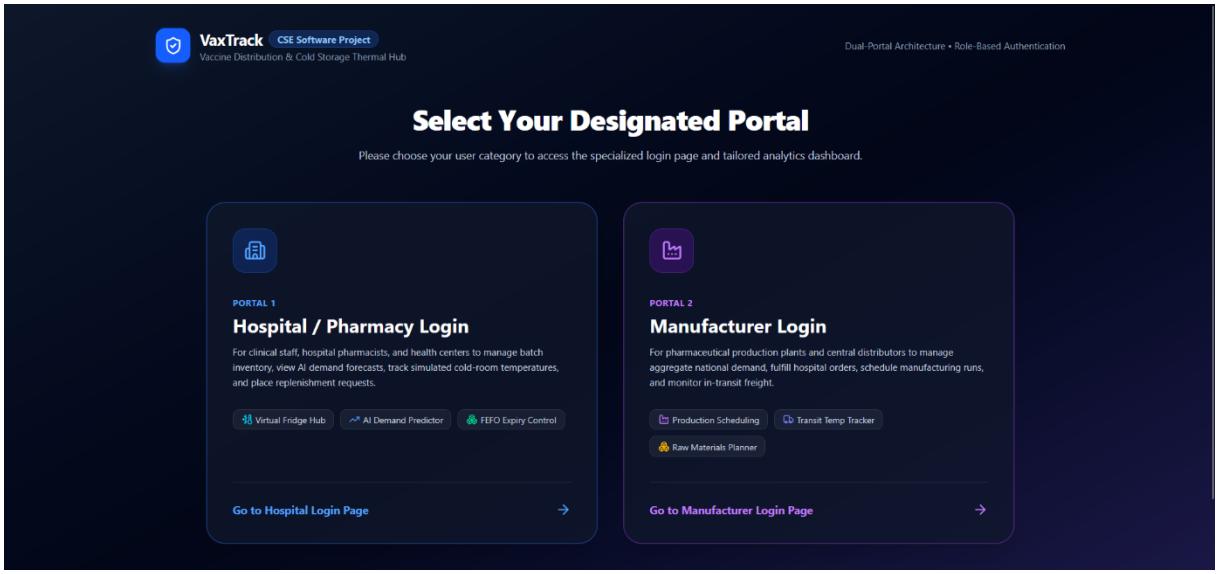
5.4: Functional Test Cases

Test ID	Test scenario	Expected result	Actual result/status
TC-01	Open the application home page	Home page loads correctly	Pending
TC-02	Log in with valid hospital/pharmacy credentials	Appropriate dashboard is displayed	Pending
TC-03	Log in with invalid credentials	Login is rejected with suitable feedback	Pending
TC-04	Attempt to access a restricted role-specific page	Access is blocked or redirected as implemented	Pending
TC-05	Enter valid inventory and batch details	Records are saved	Pending
TC-06	Submit an invalid quantity	Invalid input is rejected if validation is implemented	Pending
TC-07	Submit a supply request	Request is recorded	Pending
TC-08	Process a request as a	Request status is updated as	Pending

	manufacturer	implemented	
TC-09	Upload a valid temperature CSV	Data is processed if it meets the input requirements	Pending
TC-10	Upload a CSV with missing required columns	A suitable validation response is displayed	Pending
TC-11	Generate a temperature forecast	Forecast output is produced for valid input	Pending
TC-12	Review the stock overview	Rule-based indicators displayed	Pending

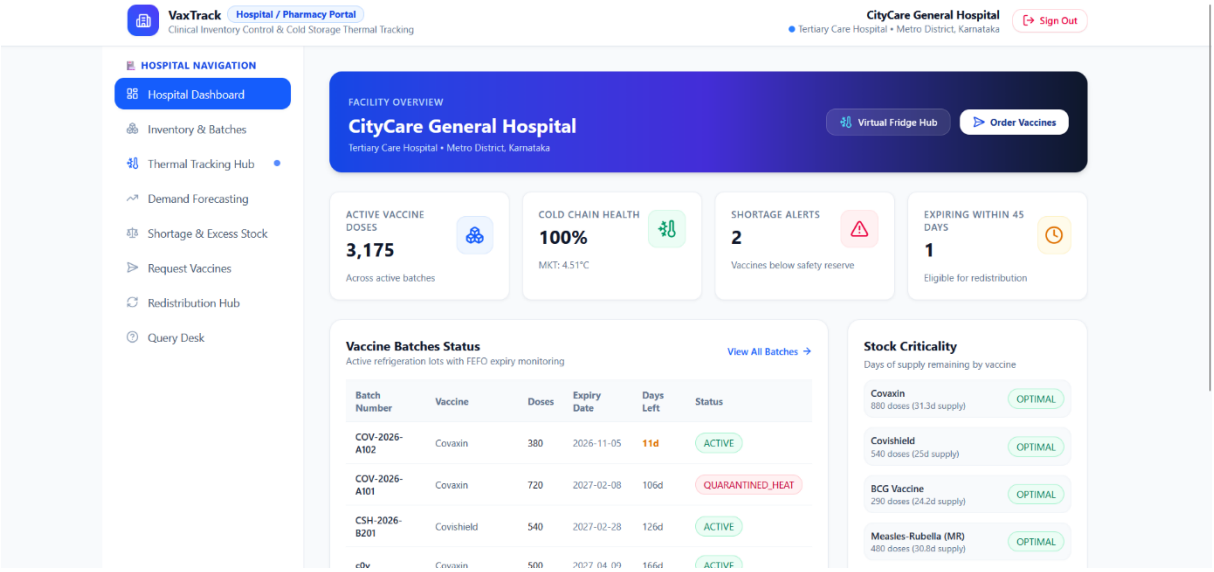
5.7 SCREENSHOT EVIDENCE

5.1: Dual-Portal Selection Interface



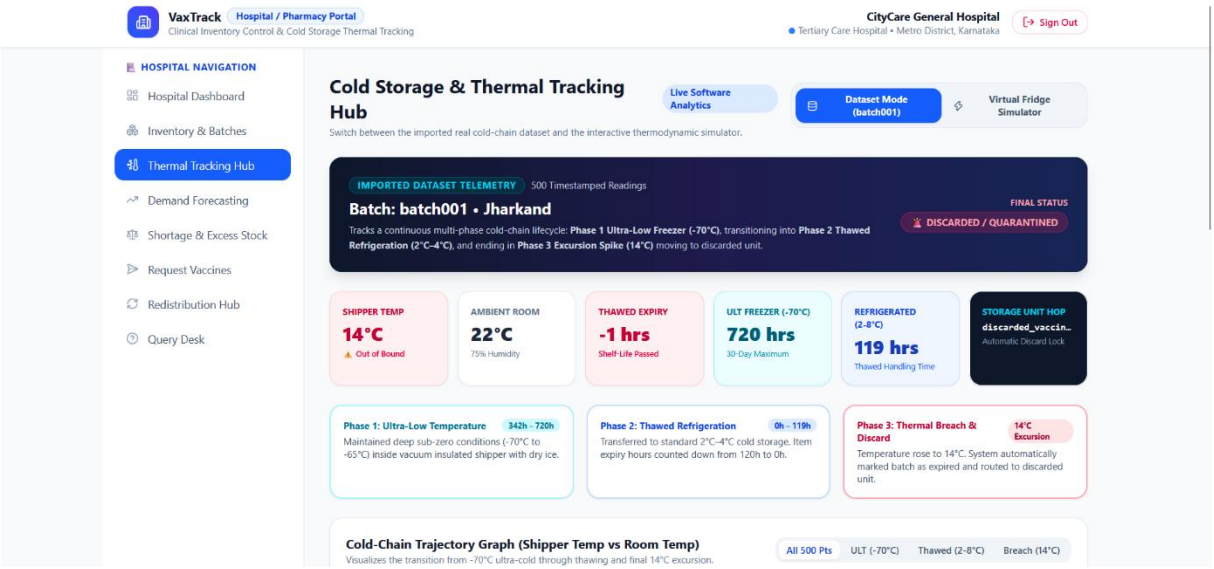
The portal selection interface provides two separate access options for the VaxTrack system: Hospital/Pharmacy and Manufacturer. Users can select the appropriate portal based on their role. This design separates hospital inventory management activities from manufacturer supply and order management activities.

5.2: Hospital Dashboard



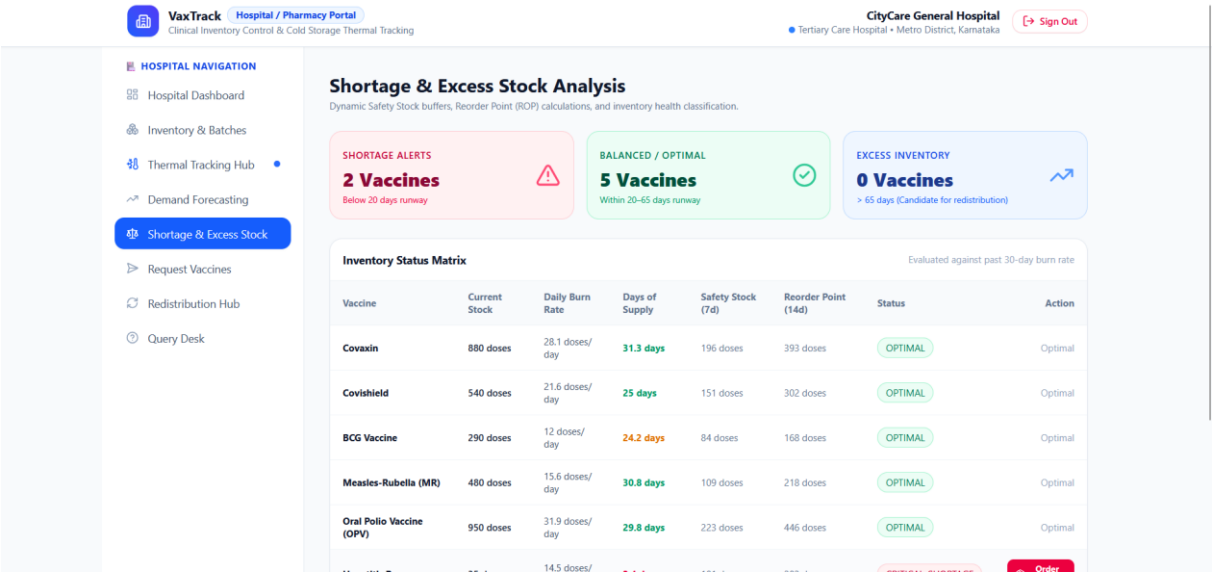
The Hospital/Pharmacy Dashboard provides an overview of vaccine inventory and distribution activities. It displays information such as active vaccine doses, cold-chain status indicators, shortage alerts, expiry details, and vaccine batch information. The dashboard helps users review stock availability and identify batches that may require attention.

5.3: Cold Storage Thermal Tracking Hub



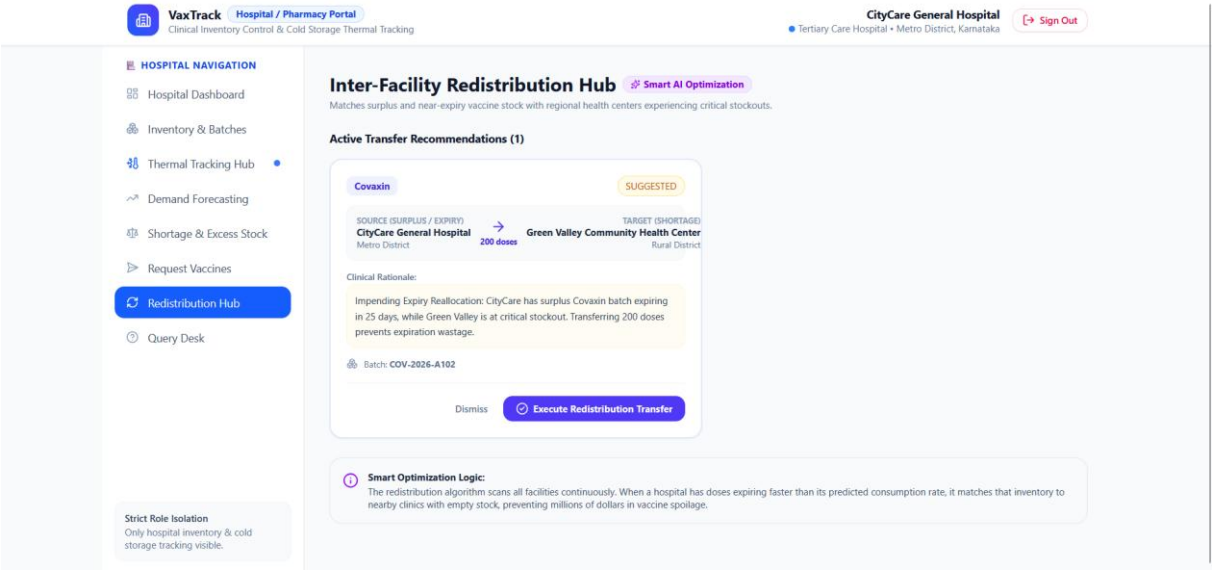
The Cold Storage Thermal Tracking Hub presents temperature information using an imported dataset or simulated readings. It displays temperature indicators, storage phases, and a cold-chain trajectory graph. The module helps demonstrate how temperature variations can be examined and how potential temperature excursions can be identified from the available data.

5.4: Shortage and Excess Stock Analysis



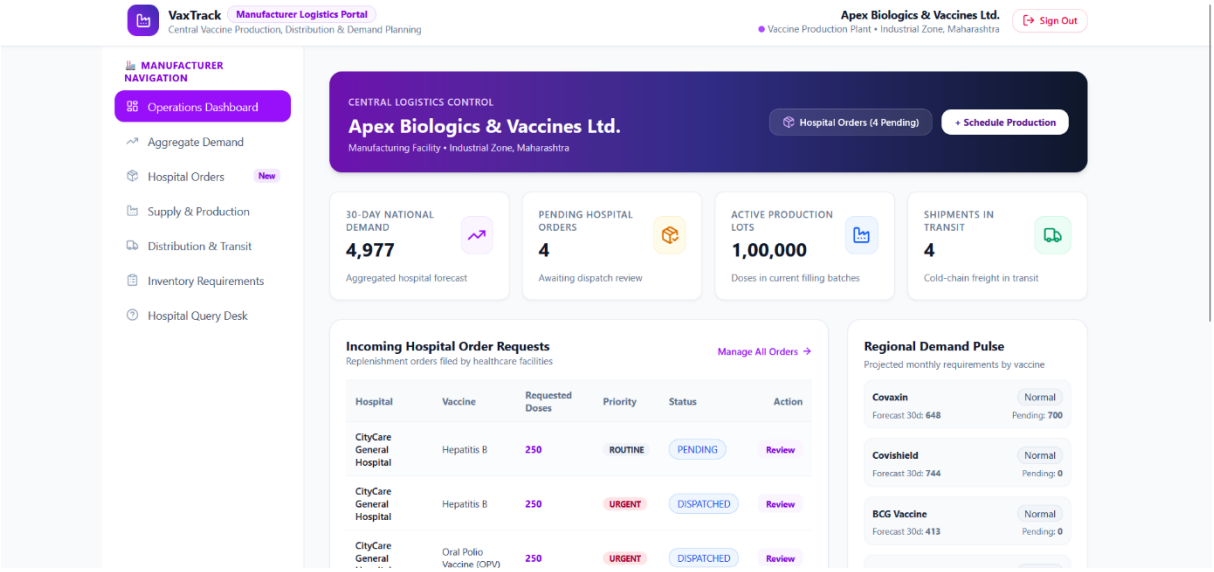
This module evaluates vaccine stock availability using inventory quantities and consumption-related values. It displays the current stock, daily usage rate, days of supply, safety stock, reorder point, and inventory status. The displayed classifications help users identify potential shortages and determine whether replenishment may be required.

5.5: Vaccine Order Request Management



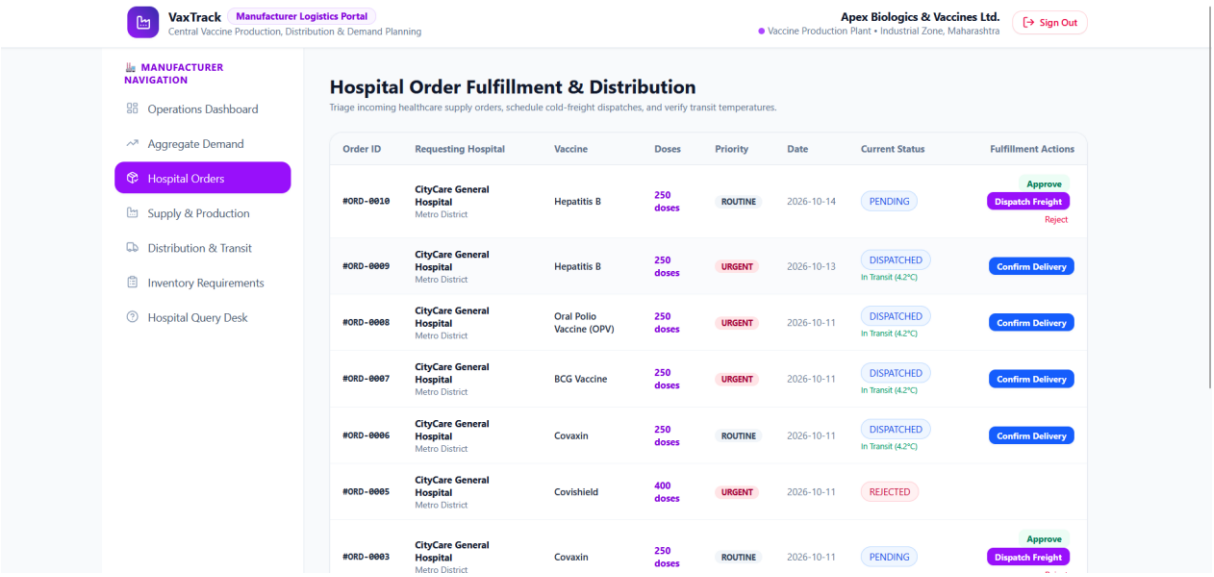
The Vaccine Order Requests module allows hospital or pharmacy users to view and manage vaccine replenishment requests. The page displays order identifiers, vaccine names, requested quantities, priority levels, manufacturers, dates, and order statuses. This information helps users track requests and review their progress.

5.6: Manufacturer Operations Dashboard



The Manufacturer Operations Dashboard provides an overview of vaccine supply and order-related activities. It displays information such as aggregated demand, pending hospital orders, production lot summaries, and shipments in transit. The dashboard helps manufacturer users review incoming requirements and access the relevant supply-management functions.

5.7: Hospital Order Fulfillment and Distribution



This module allows manufacturer users to review hospital vaccine requests and manage their displayed fulfillment status. The interface shows order details, requested quantities, priorities, dates, and current statuses. Available actions include approving, rejecting, or dispatching orders, depending on the order state and implemented workflow.

5.8 DISCUSSION

VaxTrack demonstrates how inventory records, batch and expiry information, supply requests and temperature-data analysis can be organized within a single web application. The role-based structure separates the intended tasks of hospital/pharmacy users and manufacturer users. The temperature-analysis module demonstrates the application of Random Forest regression to historical or simulated temperature data.

The project also distinguishes machine-learning functionality from rule-based logic. Temperature forecasting uses Random Forest regression, whereas stock-status and redistribution suggestions use predefined rules. This distinction is important when interpreting the system's output.

The final assessment should be based on executed functional tests, genuine screenshots and measured model-evaluation results. Performance on simulated data cannot establish performance on real cold-storage conditions, and the prototype should not be treated as a validated vaccine safety system.

5.9 STRENGTHS AND LIMITATIONS

Strengths

- Combines inventory, batch, expiry and supply request workflows.
- Provides separate workflows for hospital/pharmacy and manufacturer users.
- Demonstrates temperature forecasting using a Random Forest regression model.
- Supports repeatable software demonstrations using CSV or simulated data.
- Uses transparent rules for stock-status and redistribution indicators.
- Can be demonstrated without specialized physical hardware.

Limitations

- Does not collect live temperature readings from physical sensors.
- Does not provide real-time IoT, RFID or GPS integration.
- Forecast performance depends on dataset quality and must be measured through testing.
- Simulated data does not establish real-world cold-chain performance.
- Stock and redistribution suggestions are not trained demand-prediction or optimization models.
- The prototype is not a production-ready healthcare system.

5.5 COST–BENEFIT ANALYSIS

Item	Cost or effort	Benefit
Python and Flask	Setup and development time	Supports web application development
SQLite database	Database design and maintenance	Stores application records locally
Existing computer and browser	Use of available equipment	Enables local demonstration without additional project hardware
CSV or simulated data	Preparation and validation effort	Supports repeatable testing
Testing and documentation	Time to execute tests and capture screenshots	Improves evaluation and reproducibility
Future production hardening	Additional engineering and validation	Would be necessary before considering operational use

CHAPTER – 6
CONCLUSIONS AND FUTURE WORK

6.1 CONCLUSION

The Vaccine Distribution Inventory Control System and Cold Storage Thermal Tracking Hub (VaxTrack) is proposed to support efficient vaccine inventory management, batch and expiry tracking, distribution monitoring, and temperature data analysis. The system integrates a web-based interface with a database to maintain vaccine-related information in an organized manner. It also uses temperature datasets in CSV format or simulated data to analyse temperature readings and identify values outside configured limits.

The application incorporates a Random Forest regression approach to estimate temperature values and provides information that can support inventory review and distribution-related decisions. By organizing vaccine records and temperature information in a single system, VaxTrack aims to improve traceability, simplify record management, and support the identification of potential storage-related risks. The project provides a software-based foundation for further development of vaccine monitoring and inventory management solutions.

6.2 FUTURE ENHANCEMENTS

The system can be enhanced by integrating physical temperature sensors and IoT devices for continuous temperature monitoring. Automated alerts can be introduced to notify users when temperature readings exceed configured limits. Future versions may also incorporate advanced demand forecasting, seasonal vaccine demand analysis, and improved stock redistribution recommendations. Role-based access, detailed reports, and a more secure deployment environment can further improve the system's usability and reliability.

Overall, VaxTrack can be extended into a more comprehensive vaccine management platform by integrating real-time data collection, improved prediction techniques, and automated monitoring features.

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APPENDICES

CONTRIBUTION

SUBASRI S (7376241CS414)

- Reviewed the problem statement and related solutions.
- Helped define software-only scope and user roles.
- Reviewed inventory and expiry data fields.
- Contributed to dashboard layout and navigation.
- Prepared review slides and documented limitations.

DHISANIKA J (7376241CS170)

- Prepared and checked sample vaccine inventory data.
- Outlined missing-value and outlier handling rules.
- Defined candidate demand-prediction inputs and outputs.
- Reviewed validation metrics and test cases.
- Documented assumptions and future improvements.

GAYATHRI G(7376241CS182)

- Outlined alert conditions for low stock, expiry and simulated temperature readings.
- Designed the redistribution recommendation workflow.
- Planned dashboard tables, charts and reports.
- Prepared testing scenarios and expected results.
- Contributed to presentation and report organization.

SOURCE CODE :

1. Database Models

```
 from flask_sqlalchemy import SQLAlchemy
from datetime import datetime
db = SQLAlchemy()
class Vaccine(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(100), nullable=False)
    batch_no = db.Column(db.String(50), nullable=False)
    quantity = db.Column(db.Integer, default=0)
    expiry_date = db.Column(db.String(20))
class SupplyRequest(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    vaccine_name = db.Column(db.String(100))
    quantity = db.Column(db.Integer)
    status = db.Column(db.String(20), default="Pending")
class Temperature(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    temperature = db.Column(db.Float)
    timestamp = db.Column(db.DateTime, default=datetime.utcnow)
```

2 .Vaccine Inventory Management

```
@app.route('/inventory')
def inventory():
vaccines = Vaccine.query.all()
return render_template(
'inventory.html',
vaccines=vaccines
)
```

3 .Temperature Data Upload

```
import pandas as pd
from flask import request, render_template

@app.route('/temperature', methods=['GET', 'POST'])
def temperature():
    message = ""

    if request.method == 'POST':
        file = request.files.get('file')

        if file and file.filename.endswith('.csv'):
            data = pd.read_csv(file)

            if 'temperature' in data.columns:
                message = "Temperature data uploaded successfully."
            else:
                message = "CSV must contain a temperature column."

    return render_template(
        'temperature.html',
        message=message
    )
```

4 .Stock Status and Redistribution Suggestions

```
@app.route('/stock')
def stock():
    vaccines = Vaccine.query.all()
    suggestions = []

    for vaccine in vaccines:
        if vaccine.quantity < 10:
            status = "Low Stock"
            suggestions.append(
                f"Restock {vaccine.name
```

```
            )
        else:
            status = "Sufficient Stock"

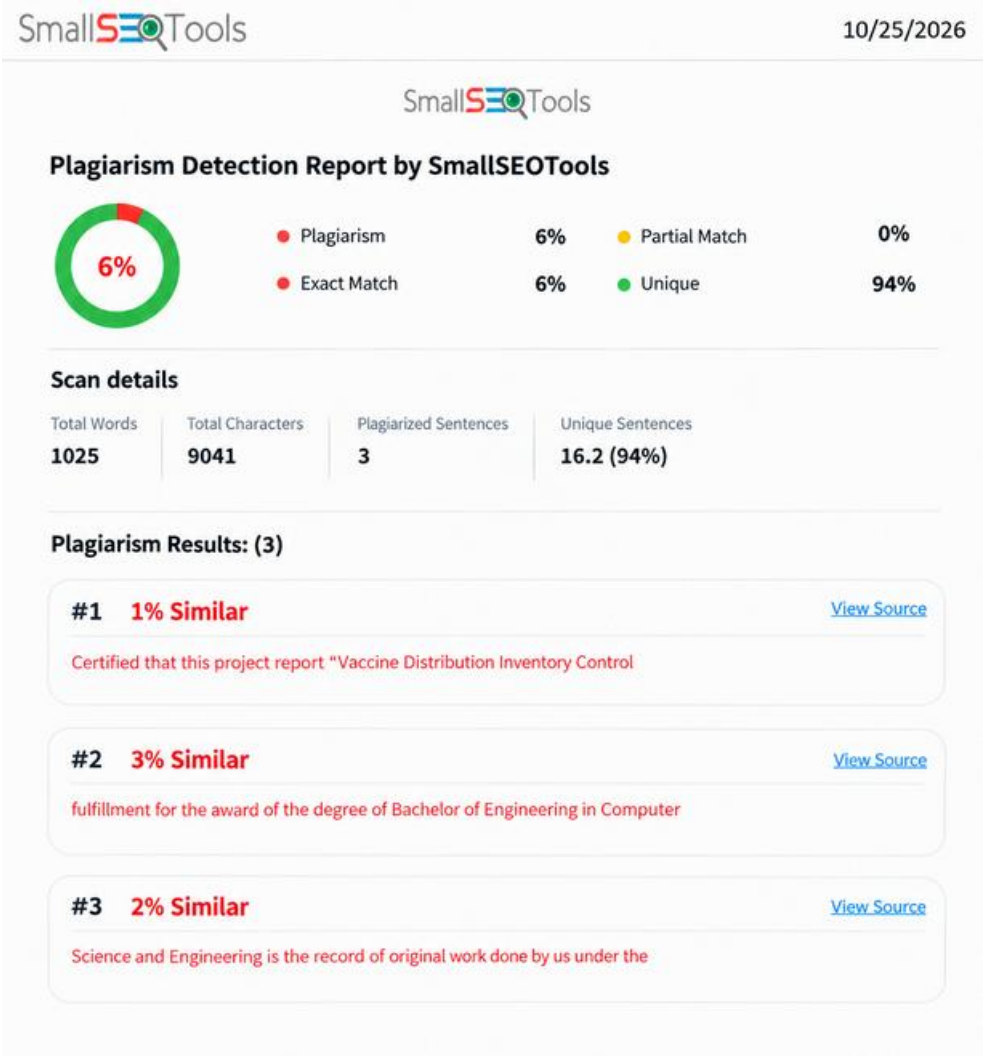
        vaccine.status = status if hasattr(
            vaccine, 'status'
        ) else status

    return render_template(
        'stock.html',
        vaccines=vaccines,
        suggestions=suggestions
    )
```

5 .Supply Request Approval and Rejection

```
@app.route('/requests/<int:item_id>/<action>')
def process_request(item_id, action):
    item = SupplyRequest.query.get_or_404(item_id)
    if action == 'approve':
        item.status = 'Approved'
    elif action == 'reject':
        item.status = 'Rejected'
    db.session.commit()
    return redirect('/requests')
```

PLAGRISM REPORT:




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